

population $X \sim N(\mu, \sigma^2)$
 sampling
 sample
 (sample value.) $\hat{\mu} =$
 $\hat{\sigma}^2 =$

Maximum likelihood estimation

$\left\{ \begin{array}{l} \text{LSE} (\hat{\beta}) \\ \text{MLE} (\hat{\mu}, \hat{\sigma}^2) \\ \text{MME} (\hat{\lambda}, \hat{\theta}) \end{array} \right.$

estimation for parameter

Joint probability distributions

Consider independent random variables $Y_1, Y_2, \dots, Y_n$ and let

$$f_i(y_i; \theta)$$

denote the probability density function if Y_i is continuous and the probability mass function if Y_i is discrete. The joint probability density function or probability mass function is then given by

$$f(\mathbf{y}; \theta) = \prod_{i=1}^n f_i(y_i; \theta).$$

Y_1, Y_2 indep $\Updownarrow$
 $f(y_1, y_2) = f(y_1)f(y_2)$

$$f(y_1, y_2, \dots, y_n; \theta) = f(\mathbf{y}; \theta) = \prod_{i=1}^n f(y_i; \theta)$$

a function of $\mathbf{y}$; θ is a parameter

Likelihood and log-likelihood

- The function

$$L(\theta; \mathbf{y}) = f(\mathbf{y}; \theta) = \prod_{i=1}^n f(y_i; \theta)$$

is called the **likelihood function**. a function of θ .

- The function

$$\begin{aligned} \ell(\theta; \mathbf{y}) = \log L(\theta; \mathbf{y}) &= \log \prod_{i=1}^n f(y_i; \theta) \\ &= \sum_{i=1}^n \log (f(y_i; \theta)) \end{aligned}$$

is called the **log-likelihood function**.

Example 5.1

Suppose $y_1, y_2, \dots, y_n$ are *i. i. d.* $Po(\lambda)$ observations.

(a) Give the likelihood function.

(b) Give the log-likelihood function.

$$E(y_i) = \text{Var}(y_i) = \lambda.$$

$$f(y_i; \lambda) = \frac{e^{-\lambda} \lambda^{y_i}}{y_i!} \quad y_i = 0, 1, 2, \dots$$

$$(a) \quad L(\lambda; \mathbf{y}) = \prod_{i=1}^n \frac{e^{-\lambda} \lambda^{y_i}}{y_i!} = \frac{e^{-n\lambda} \lambda^{\sum_{i=1}^n y_i}}{\prod_{i=1}^n y_i!}$$

$$(b) \quad l(\lambda; \mathbf{y}) = \log \left(\frac{e^{-n\lambda} \lambda^{\sum y_i}}{\prod y_i!} \right) = -n\lambda + \sum y_i \log \lambda - \sum \log(y_i!)$$

Example 5.2

Suppose $y_1, y_2, \dots, y_n$ are *i. i. d.* $N(\mu, \sigma^2)$ observations with σ^2 known.

- (a) Give the likelihood function.
- (b) Give the log-likelihood function.

$$f(y_i; \mu) = \frac{1}{\sqrt{2\pi}\sigma} e^{-\frac{(y_i - \mu)^2}{2\sigma^2}}$$

$$\begin{aligned} (a) \quad L(\mu; \mathbf{y}) &= \prod_{i=1}^n \frac{1}{\sqrt{2\pi}\sigma} e^{-\frac{(y_i - \mu)^2}{2\sigma^2}} \\ &= (2\pi\sigma^2)^{-\frac{n}{2}} e^{-\frac{\sum (y_i - \mu)^2}{2\sigma^2}} \end{aligned}$$

$$(b) \quad l(\mu; \mathbf{y}) = -\frac{n}{2} \log(2\pi\sigma^2) - \frac{\sum (y_i - \mu)^2}{2\sigma^2}$$

if μ and σ^2 unknown

$$L(\mu, \sigma^2; \mathbf{y}) = (2\pi)^{-\frac{n}{2}} (\sigma^2)^{-\frac{n}{2}} e^{-\frac{1}{2\sigma^2} \sum (y_i - \mu)^2}$$

$$l(\mu, \sigma^2; \mathbf{y}) = -\frac{n}{2} \log 2\pi - \frac{n}{2} \log(\sigma^2) - \frac{1}{2\sigma^2} \sum_{i=1}^n (y_i - \mu)^2$$

Score function

If $y_1, y_2, \dots, y_n$ are independent observations with log-likelihood $\ell(\theta; \mathbf{y})$, then the function

$$S(\theta; \mathbf{y}) = \frac{\partial \ell}{\partial \theta}$$

is called the **score function**.

score vector.

$$\theta = (\theta_1, \theta_2)$$

$$S(\theta; \mathbf{y}) = \begin{bmatrix} \frac{\partial \ell}{\partial \theta_1} \\ \frac{\partial \ell}{\partial \theta_2} \end{bmatrix}$$

Example 5.3

Find the score function if $y_1, y_2, \dots, y_n$ are i. i. d. with the following distributions:

(a) $Po(\lambda)$. $l(\lambda; y) = -n\lambda + \sum y_i \log \lambda + \log \left[\prod_{i=1}^n \frac{1}{y_i!} \right]$

(b) $N(\mu, \sigma^2)$ with σ^2 known.

$$l(\mu; y) = -\frac{n}{2} \log(2\pi\sigma^2) - \frac{1}{2\sigma^2} \sum (y_i - \mu)^2$$

$$a) S(\lambda; y) = \frac{\partial l(\lambda; y)}{\partial \lambda} = -n + \frac{\sum y_i}{\lambda}$$

$$\begin{aligned} b) S(\mu; y) &= \frac{\partial l(\mu; y)}{\partial \mu} = +\frac{1}{2\sigma^2} 2 \sum (y_i - \mu) \\ &= \frac{1}{\sigma^2} \sum (y_i - \mu) \end{aligned}$$

Maximum likelihood estimation

If $y_1, y_2, \dots, y_n$ are independent observations with log-likelihood $\ell(\theta; \mathbf{y})$, then the **maximum likelihood estimate** (MLE) $\hat{\theta}$ is the value of θ that maximizes $\ell(\theta; \mathbf{y})$.

$$\hat{\theta} = \underset{\theta}{\operatorname{argmax}} \ell(\theta; \mathbf{y}) \quad \hat{\theta}: \max \ell(\theta, \mathbf{y}) \text{ 时 } \theta?$$

$$= \underset{\theta}{\operatorname{argmax}} \ell(\theta; \mathbf{y})$$

$$S(\theta; \mathbf{y}) = \frac{\partial \ell(\theta; \mathbf{y})}{\partial \theta} = 0$$

$$\hat{\theta} = ?$$

Maximum likelihood estimation

In practice, $\hat{\theta}$ is usually derived by solving the score equation

$$S(\theta; \mathbf{y}) = 0.$$

We assume $\hat{\theta}$ exists and is unique.

Example 5.4

$$E(y_i) = \underline{\lambda}$$

Suppose $y_1, y_2, \dots, y_n$ are *i. i. d.* $Po(\lambda)$ observations.
Find the maximum likelihood estimate $\hat{\lambda}$ of λ .

$$S(\lambda; \mathbf{y}) = -n + \frac{1}{\lambda} \sum y_i$$

$$-n + \frac{1}{\lambda} \sum y_i = 0$$

$$\frac{1}{\lambda} \sum y_i = n$$

$$\hat{\lambda}_{MLE} = \frac{1}{n} \sum y_i = \bar{y}$$

$$E(\hat{\lambda}_{MLE}) = E\left(\frac{1}{n} \sum y_i\right) = \frac{1}{n} n E(y_i) = \lambda$$

Example 5.5

Suppose $y_1, y_2, \dots, y_n$ are *i. i. d.* $N(\mu, \sigma^2)$ observations with σ^2 known. Find the maximum likelihood estimate $\hat{\mu}$ of μ .

$$S(u, \lambda) = \frac{1}{\sigma^2} \sum (y_i - u) = 0$$

$$\sum (y_i - u) = 0$$

$$\sum y_i - nu = 0$$

$$\hat{\mu}_{MLE} = \frac{\sum y_i}{n} = \bar{y}$$

Example 5.6

Suppose $y_1, y_2, \dots, y_n$ are *i. i. d.* $N(\mu, \sigma^2)$ observations **where both μ and σ^2 are unknown**. Find the maximum likelihood estimate $\hat{\mu}$ and $\hat{\sigma}^2$ of μ and σ^2 , respectively.

$$\begin{aligned} S(\sigma^2; y) &= \frac{\partial}{\partial \sigma^2} \ell(\mu, \sigma^2; y) \\ &= -\frac{n}{2} \frac{1}{\sigma^2} + \frac{1}{2\sigma^4} \sum (y_i - \mu)^2 \end{aligned}$$

score vector

$$S(\mu, \sigma^2; y) = \begin{bmatrix} \frac{1}{\sigma^2} \sum (y_i - \mu) \\ -\frac{n}{2\sigma^2} + \frac{1}{2\sigma^4} \sum (y_i - \mu)^2 \end{bmatrix}$$

$$\begin{aligned} -\frac{n}{2\sigma^2} + \frac{1}{2\sigma^4} \sum (y_i - \hat{\mu})^2 &= 0 \\ \frac{1}{2\sigma^4} \sum (y_i - \hat{\mu})^2 &= \frac{n}{2\sigma^2} \end{aligned}$$

Example 5.6 Solutions

$$n \hat{\sigma}^2 = \sum_{i=1}^n (y_i - \hat{u})^2$$
$$\hat{\sigma}^2_{MLE} = \frac{\sum (y_i - \hat{u})^2}{n}$$